II. Fill in the blanks:

12. The number of all permutations of n distinct things taken all at a time is _____

13. Number of ways of 4 students can stand in a queue is _____

14. From a group of 10 students 2 students are selected in _____ ways

15. The value of nC_n is equal to _____

16. The value of $21C_3$ is equal to _____

17. $\lim_{x \to 5} \frac{2x+3}{x+5}$ is equal to _____

18. The derivative of e^x is equal to _____

19. The integral of $\frac{1}{x}$ is _____

20. The value of $\int 6x^5\, dx$ is _____

21. The value of $\int_0^1 x^{10}\, dx$ is _____

III . Very Short Answer Questions :

22. A boy has 6 pants and 10 shirts. In how many ways can he wear them?

23. Evaluate (i) $4P_4$ (ii) $10P_4$ (iii) $100P_2$

24. How many different four letter code words can be formed with the letter 'SWIPE' no letter is not repeated.

25. In a competition, in how many ways can first and second place be awarded to 10 people?

26. Evaluate $10C_4$, $22C_3$, $100C_{98}$

27. Evaluate (i) $\lim_{x \to 2} \frac{x^2-4x+7}{x+8}$ (ii) $\lim_{x \to 5} \frac{x^2-25}{x-5}$

28. Find the derivative of x^2e^x

IV. Short Answer Questions :

29. How many different words can be formed with letters of the word 'PROBABILITY'?

30. A bag contains 7 blue balls and 5 red balls. Determine the number of ways in which 3 blue and 2 red balls can be selected.